

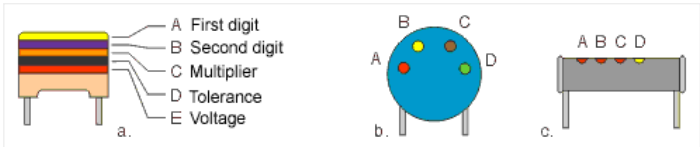
## 2.1.1 MARKING THE BLOCK-CAPACITORS

### MIKROELEKTRONIKA

Commonly, capacitors are marked by a set of numbers representing the capacity. Beside this value is another number representing the maximal working voltage, and sometimes tolerance, temperature coefficient and some other values are printed as well. But on the smallest capacitors (such as surface-mount) there are no markings at all and you must not remove them from their protective strips until they are needed. The size of a capacitor is never an indication of its value as the dielectric and the number of layers or “plates” can vary from manufacturer to manufacturer. The value of a capacitor on a circuit diagram, marked as 4n7/40V, means the capacitor is 4,700pF and its maximal working voltage is 40v. Any other 4n7 capacitor with higher maximal working voltage can be used, but they are larger and more expensive.

Sometimes, capacitors are identified with colors, similar to the 4-band system used for resistors (figure 2.2). The first two colors (A and B) represent the first two digits, third color (C) is the multiplier, fourth color (D) is the tolerance, and the fifth color (E) is the working voltage.

With disk-ceramic capacitors (figure 2.2b) and tubular capacitors (figure 2.2c) working voltage is not specified, because these are used in circuits with low DC voltage. If a tubular capacitor has five color bands on it, the first color represents the temperature coefficient, while the other four specify the capacity in the previously described way.



| COLOR  | DIGIT | MULTIPLIER | TOLERANCE | VOLTAGE |
|--------|-------|------------|-----------|---------|
| Black  | 0     | x 1 pF     | ±20%      |         |
| Brown  | 1     | x 10 pF    | ±1%       |         |
| Red    | 2     | x 100 pF   | ±2%       | 250V    |
| Orange | 3     | x 1 nF     | ±2.5%     |         |
| Yellow | 4     | x 10 nF    |           | 400V    |
| Green  | 5     | x 100 nF   | ±5%       |         |
| Blue   | 6     | x 1 µF     |           |         |
| Violet | 7     | x 10 µF    |           |         |
| Grey   | 8     | x 100 µF   |           |         |
| White  | 9     | x 1000 µF  | ±10%      |         |

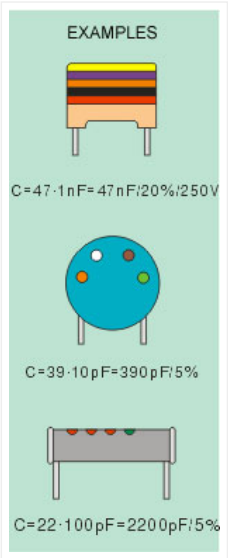
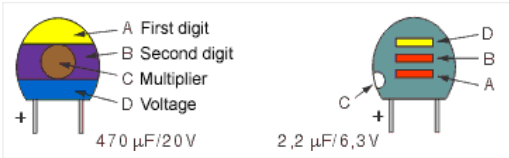


Fig. 2.2: Marking the capacity using colors

The figure 2.3 shows how the capacity of miniature tantalum electrolytic capacitors are marked by colors. The first two colors represent the first two digits and have the same values as with resistors. The third color represents the multiplier, to get the capacity expressed in  $\mu\text{F}$ . The fourth color represents the maximal working voltage.



| COLOR  | DIGIT | MULTIPLIER          | VOLTAGE |
|--------|-------|---------------------|---------|
| Black  | 0     | x 1 $\mu\text{F}$   | 10V     |
| Brown  | 1     | x 10 $\mu\text{F}$  |         |
| Red    | 2     | x 100 $\mu\text{F}$ |         |
| Orange | 3     |                     |         |
| Yellow | 4     |                     | 6.3V    |
| Green  | 5     |                     | 16V     |
| Blue   | 6     |                     | 20V     |
| Violet | 7     |                     |         |
| Grey   | 8     | x .01 $\mu\text{F}$ | 25V     |
| White  | 9     | x .1 $\mu\text{F}$  | 3V      |
| Pink   |       |                     | 35V     |

Fig. 2.3: Marking the tantalum electrolytic capacitors

One important note on the working voltage: The voltage across a capacitor must not exceed the maximal working voltage as the capacitor may get destroyed. In the case when the voltage is unknown, the “worst” case should be considered. There is the possibility that, due to malfunction of some other component, the voltage on capacitor equals the power supply voltage. If, for example, the supply is 12V, the maximal working voltage for the capacitor should be higher than 12V.

